

Chen Xie

Department of Physics & Astronomy

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Research Position

- **Assistant Research Scientist** Baltimore, MD
The Johns Hopkins University 2023 - present
- Graduate Student Researcher Marseille, France
Aix-Marseille Université 2020 - 2023
- Summer Student internship Dwingeloo, The Netherlands
ASTRON/JIVE 2017

Education

- **Aix-Marseille Université** Marseille, France
PhD in Astronomy awarded in Feb 2024
 - PhD Thesis: Development of advanced post-processing methods for the direct detection of exoplanets with ground-based high-contrast imagers
 - Thesis Advisor: Dr. Arthur Vigan, Dr. Elodie Choquet
- **Leiden University** Leiden, The Netherlands
M.Sc. in Astronomy Feb 2020
 - Major Thesis: VLT/MUSE as a high-contrast imager
 - Thesis Advisors: Prof. Matthew Kenworthy, Dr. Jos de Boer, and Dr. Sebastiaan Haffert
 - Minor Thesis: Radio Observations of Frontier Fields Clusters Abell S1063 and Abell 370: The Discovery of New Radio Halos
 - Thesis Advisor: Prof. Reinout van Weeren
- **Xiamen University** Xiamen, China
B.Sc. in Physics June 2016
 - Bachelor Thesis: On the Host Galaxy of GRB 150101B
 - Thesis Advisor: Prof. Taotao Fang

Selected Grants & Awards

Total awards for JWST programs as Grant Admin PI: **\$149,369** Cycle 3 – 4
Poster Prize, Lyot conference 2022
Oort Scholarship (**€22,000 + tuition waiver**), Leiden University 2018 – 2020
First Prize of Lin Qiao Prize for Excellent Undergraduate Research Project, Peking University 2016

Mentoring

6. Luo, J.-S. (BSc student, Xiamen Univ., ongoing) – SAFFORN survey, spiral motion measurement
5. Chen, J.-Y. (MSc student, Xiamen Univ., ongoing) – SAFFORN survey
4. Zhong, H.-S. (PhD student, Sun Yat-sen Univ., ongoing) – Reflectance spectra of HD 32297 debris disk; paper in prep.
3. Xie, C.-Y. (PhD student, Univ. of Arizona, 2024) – publication in *Universe*, 10, 465 (2024).
2. Ahmad, Y. (MSc student, EuroPhotonics, 2022–23) – developing image-processing techniques

1. Rahman, S. (MSc student, EuroPhotonics, 2021–22) – developing image-processing techniques

Publications (148 citations from 9 first-/corresponding-author papers)

Total 20 papers, incl. 1 in Nature (1st authored), 1 in Science (3rd, in rev.), 1 in Nature Astronomy (5th)

1st-/Corresponding-authored:

1. **Xie, C.**, et al. *Water ice in the debris disk around HD 181327*, **Nature**, 641, 608 (2025).
Research Highlight: water ice discovery
2. **Xie, C.**, et al. *Reference-star differential imaging on SPHERE/IRDIS*, A&A, 666, A32 (2022).
Research Highlight: new archival RDI with $2.1\times$ improvement
3. **Xie, C.**, et al. *Searching for proto-planets with MUSE*, A&A, 644, A149 (2020).
Research Highlight: first MUSE survey
4. **Xie, C.**, et al. *DESTINYs: Dynamical evidence of a spiral-arm-driving protoplanet and Gap-Opening Protoplanet from SAO 206462 Spiral Motion*, Universe, 10, 465 (2024).
5. Ren, B.B., **Xie, C.**, et al. *A companion in V1247 Ori supported by motion in the pattern of the spiral arm in a protoplanetary disk*, A&A, 681, L2 (2024).
6. **Xie, C.**, et al. *Dynamical detection of a companion driving a spiral arm in a protoplanetary disk*, A&A, 675, L1 (2023).
7. **Xie, C.**, et al. *A MUSE view of the asymmetric jet from HD 163296*, A&A, 650, L6 (2021).
8. **Xie, C.**, et al. *The discovery of radio halos in the Frontier Fields clusters Abell S1063 and Abell 370*, A&A, 636, A3 (2020)
9. **Xie, C.**, et al. *On the Host Galaxy of GRB 150101B and the Associated Active Galactic Nucleus*, ApJL, 824, L17 (2016)

2nd/3rd authored:

10. Betti, S.K., Chen, C.H., **Xie, C.**, et al. *Detection of ices in the debris disk around β Pictoris*, **Science** (under review since Feb. 2025).
11. Matthews, E.C., Bonnefoy, M., **Xie, C.**, et al. *The first scattered light images of HD 112810, a faint debris disk in the Sco-Cen association*, A&A, 679, A58 (2023).
12. Uyama, T., **Xie, C.**, et al. *Keck/OSIRIS Pa β High-contrast Imaging and Updated Constraints on PDS 70b*, AJ, 162, 214 (2021).

Contributing authored:

13. Worthen, K., Chen, C.H., Brittain, S.D., Najita, J.R., **Xie, C.** et al. *Fluorescently excited CO emission in the 49 Ceti debris disk spatially resolved by JWST/NIRSpec*, **Nature Astronomy**, 9, 1680 (2025)
14. Zhong, H.-S., Ren, B., Ma, B., **Xie, C.** et al. *Multiband reflectance and shadowing of the protoplanetary disk RX J1604.3-2130 in scattered light*, A&A, 684, A168 (2024)
15. Lu, C.X. et al. (incl. **Xie, C.**) *JWST/NIRSpec Detects Warm CO Emission in the Terrestrial-Planet Zone of HD 131488*, accepted to ApJ
16. Engler, N. et al. (incl. **Xie, C.**) *Characterization of debris disks observed with SPHERE*, A&A, 704, A21 (2025).

White papers (non peer-reviewed):

17. **Xie, C.** et al. *The transmission spectrum of a terrestrial planet-forming zone*, Roman Coronagraph White Papers (2025).
18. Chen, C.H., et al. (incl. **Xie, C.**). *Water Delivery in the η Crv Habitable Zone*, Roman Coronagraph White Papers (2025).
19. Lewis, B.L., et al. (incl. **Xie, C.**). *Characterizing Known Debris Disk HR 4796A in Polarized Scattered Light*, Roman Coronagraph White Papers (2025).
20. Hom, J., et al. (incl. **Xie, C.**). *A First Look at Bright Debris Disks with HLC Band 1*, Roman Coronagraph White Papers (2025).

Leadership

Spiral motion lead, VLT/SPHERE SAFFRON survey (PI: B. Ren; 79 hr; 40+ members), primary goal

Selected Meetings since 2022

STScI-JHU ExoJamboree, **Talk**, Baltimore, MD, 2025
 Gordon Research Conferences, Origins of Solar Systems, Poster, South Hadley, MA, 2025
 National Capital Area Disks Meeting, **Talk**, Washington, DC, 2025
 STScI JWST science conversation, **Invited talk**, Baltimore, MD, 2025
 Earth & Planets Laboratory Seminar, **Invited talk**, Washington, DC, 2025
 STScI-JHU ExoJamboree, **Talk**, Baltimore, MD, 2024
 Dust Devils workshop, **Talk**, Tucson, AZ, 2024
 AO4ELT, Poster, Marseille, 2023
 Observatory of the Côte d’Azur Seminar, **Invited talk**, Nice, 2023
 ETH Zürich Exoplanets & Habitability Seminar, **Invited talk**, Zürich, 2022
 Observatory of the Côte d’Azur Seminar, **Invited talk**, Nice, 2022
 Debris discs: At Home and Abroad, **Talk**, Jena, 2022
 Spirit of Lyot 2022, Poster, Leiden, 2022

Service & Outreach

Member, Roman Coronagraph Community Participation Program 2025 –
 Referee, AAS journals 2024 –
 Referee, A&A 2023 –
 Organizer, sidewalk astronomy on XMU campus (in Chinese) 2014 – 2017
 Organizer & presenter, exhibition of the amateur astronomer club in XMU (in Chinese) 2015 – 2016

Press & Media

NASA | Another First: NASA Webb Identifies Frozen Water in Young Star System 2025
 BBC | This baby star is hiding water ice, and maybe the seeds of a new earth 2025
 Space | James Webb Space Telescope discovers an alien planetary system’s icy edge . . . 2025
 APOD | HD 163296: Jet from a Star in Formation 2021

Successful Telescope Proposals

Total: 37.7 hr (JWST), 4 orbits (HST), 73.5 hr (VLT), 5 hr (VLTI), 1.5 nights (Subaru), 23.6 hr (radio),
 incl. 8.9 hr (JWST, US Admin PI), 2 hr (GRAVITY, PI), 3 hr (MUSE, d-PI), 15.6 hr (radio, PI)

- JWST (GO 6991): Testing planet-disk interaction theory by imaging a unique protoplanet
Admin-PI, 8.9 hr, \$106,069 2025
- HST (GO 18026): Testing volatile transportation in the epoch of terrestrial planet formation
Col-I, 4 orbits 2025
- JWST (GO 7313): Determining the Origin of the Gas in the 49 Ceti Debris Disk
Col-I, 7.2 hr 2025
- JWST (GO 6940): Determining the Origin of Water Ice in the β Pictoris Disk
Col-I, 13.7 hr 2025
- VLTI/GRAVITY: Pinning down the orbit of HD 100453 B
PI, 1 hr 2025
- Subaru: Confirming the direct detection of a runaway accreting protoplanet
Co-I, 1 night 2025
- JWST (GO 5261): Confirming the youngest gap-opening protoplanet
Col-I, 7.9 hr, \$43,300 2024
- VLT/SPHERE: Locating hidden giant planets: the SAFFRON survey
Co-I, 15 hr 2024
- VLTI/GRAVITY: Pinning down the orbit of HD 100453 B
PI, 1 hr 2024
- VLTI/GRAVITY: Confirming the youngest gap-opening protoplanet
Co-I, 3 hr 2024
- VLT/VISIR: Do exocomets release dust?
Co-I, 1 hr 2023
- VLT/SPHERE: Locating hidden giant planets: the SAFFRON survey
Co-I, 54.5 hr 2023
- VLT/MUSE: Astrochemistry hints at the presence of a young accreting planet
d-PI, 3 hr 2022
- Subaru: Astrochemistry hints at the presence of a young accreting planet
Co-I, 0.5 night 2022
- GMRT: Testing turbulent re-acceleration in Abell S1063
PI, 10 hr 2019
- VLA: Testing turbulent re-acceleration in Abell S1063
PI, 5.6 hr 2019
- LOFAR: Observing the previously undetected diffuse radio emission
Col-I, 8 hr 2017
- CFHT: Studying the origin of X-ray arcs in M51b
Col-I, 1 hr 2016